SANKUR KUNDU

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10 fixes merged: NVIDIA, Apple, Meta, CMU, Oxford, Hugging Face • Harvard textbook lab • NASA CARA review

OPEN-SOURCE CONTRIBUTIONS: TEN FIXES MERGED IN 18 DAYS, 16 OCT – 2 NOV 2026

NVIDIA ISING-DECODING | NVIDIA's AI decoders for quantum error correction | #98, #99 **Merged 30 Oct & 2 Nov 2026**
- **Two fixes merged:** NVIDIA's published Hugging Face models now load in its tests, and both checkpoint settings work as documented; maintainer Ivan Basov: "thank you for the contribution and for working through the review rounds."

APPLE MLX | Apple's ML framework for Apple silicon, **29k★** | PR #4737 **Merged 22 Oct 2026**
- Fixed a data race in MLX's distributed backend (ThreadSanitizer: 2–4 races per run → **0**); merged by MLX maintainer @zcbenz.

META EXECUTORCH | On-device AI in Instagram & WhatsApp, **5.1k★** | #23677, #23912 **Merged 16 & 28 Oct 2026**
- **Two fixes merged** into Meta's on-device LLM kernels: closed an out-of-bounds read in the 2/4-bit embedding kernels and fixed checks aborting valid fp16-scale models; approved by Meta's Scott Roy ("LGTM! Thanks for the fix!") and Jacob Stevens.

CMU XGRAMMAR | Structured output in vLLM, SGLang & NVIDIA TensorRT-LLM | PR #1012 **Merged 29 Oct 2026**
- Fixed jump-forward decoding crashing whenever a grammar forces part of a UTF-8 character, a bug open since issue #132.

HUGGING FACE OPENENV | RL environments (Meta-PyTorch, NVIDIA, Microsoft), **2.8k★** | #1441, #1442 **Merged 19 Oct 2026**

JAXMARL | NeurIPS 2024; BOLD Lab: Oxford, UCL & Imperial, **868★** | PR #206 **Merged 26 Oct 2026**
- Fixed rollouts behind 8 Q-learning baselines that zeroed each episode's first observation; lead author Alex Rutherford: "Good find!!"

OXFORD PYZX | Quantum-circuit optimiser, **553★** | PR #563 **Merged 22 Oct 2026**

In review: Google JAX (assigned to co-author M. Johnson), vLLM (**95.7k★**), SGLang (**38.4k★**), PyTorch, MIT, ETH Zurich, IBM

HONORS & RECOGNITION

- **Harvard's ML Systems textbook** (Prof. Vijay Janapa Reddi): **built and submitted a full lab** on why temperature-0 LLMs are not deterministic (**PR #2175**), passing the book's test suites (**2,658 tests, 0 failures**); team: "the feature looks great" (discussion).
- **NASA CARA** (NASA's conjunction-assessment team): my reproduction of its Monte Carlo collision probabilities (**53/53** cases, **15–19×** faster per core) is **under review by NASA CARA**, referred by its Operations Team Lead (emails).
- **Entrepreneur First** (UK): scouted me from public work, citing my cuBLAS-parity GEMM and a negative result: "which I rarely see".

RESEARCH (SOLE AUTHOR)

- **"Removing the Token Tax from an Existing Model"**, **"The Sampler Was Not the Policy"**, **"Reproducing NASA CARA's Monte Carlo Probability of Collision"** (arXiv, forthcoming).

PROJECTS

ratchet | PyTorch, C++/CUDA: RL post-training (GRPO) from scratch | study page
- Lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** in 100 steps on 2× T4, scoring all 1,319 test problems.
- Cut weight sync **369×** with in-place GPU updates, and made steps **1.9×** faster by overlapping generation with training.
- Found fp16 weight sync silently erasing most updates, so the sampler ran a policy the trainer never had (gap **5.2** → **0.04** nats).

tessel | Rust: a tile language and compiler for GPU and TPU kernels | study page
- Matched **cuBLAS** on GEMM (**1.00–1.04×**) and beat PyTorch SDPA on causal attention (**1.3–1.9×**) on a T4, checked against fp64.
- Built an LLM engine on these kernels alone that matches **vLLM token for token at 1.05× its throughput** (TinyLlama-1.1B).

lipi | PyTorch: cutting the token tax on Indian languages | study page
- Built a tokenizer that needs fewer tokens than **GPT-4o's and Gemini's in 23 of 24** Indian languages (on Hugging Face).
- Extended Qwen2.5-0.5B with it: Odia takes **8.6×** fewer tokens and, in equal GPU time, is predicted **20% better**.

samebit | C, CUDA, PyTorch, MLX: LLM inference in integers, the same bits on every chip
- Ran **Qwen2.5-0.5B (494M parameters) entirely in integers**: within **0.75%** of float32 perplexity and **bit-identical on an NVIDIA T4 GPU and a CPU**.
- **Six backends** (C, NumPy, PyTorch, **MLX**, CUDA) give one digest; its lab is **submitted to Harvard's ML Systems textbook**.

More projects
- **chirp**: GPU gravitational-wave search built from general relativity; finds **62 of 71** LIGO O4a events blind, all 71 at **95%** of catalog SNR.
- **ink-sensitivity**: official Vesuvius ink model finds **79%** of real ink moved into an unread scroll (**AUC 0.94**).
- **relay**: fault-tolerant LLM serving on custom kernels, prefill **8.7×** faster, **384/384** requests identical through 60 kills.
- **syndrome**: GPU quantum decoder bit-identical to **IBM's**, **28.7×** faster per shot; **kessler**: matches **NASA CARA** on 53/53.

EXPERIENCE

GIGZEN | Co-founder & Founding Engineer | Bhubaneswar, India **Apr 2026 – Present**
Early-stage startup (name approved by India's MCA; incorporation in progress) | Sole engineer
- Built Waggle, three delivery apps on one PostgreSQL backend: **53 tables, 90 RLS policies, 154 SQL functions**; **150 attacks** blocked.
- Released **Populace**, an open-source testing tool (**1,600+ npm downloads**) that found **5 bugs** QA missed.

TECHNICAL SKILLS

ML systems: PyTorch, JAX, MLX; GRPO, DDP, ZeRO/FSDP, MoE expert parallelism, paged KV cache, int8 quantization
Languages & GPU: Python, C++20, CUDA (tensor cores), Rust, C, Swift, SQL; Triton-style kernels, Metal, Pallas (TPU)

EDUCATION

IEM KOLKATA | B.Tech, Computer Science and Engineering (AI & ML) | Kolkata, India **Sep 2022 – Apr 2026**